

MSc. Thesis: Android Secure WiFi

Initial Connection with GeoPKI

When initially connecting to a WiFi network (or any other over-the-air network), a user typically (like with the EDUROAM WiFi) obtains a certificate used to verify the identity of the login (e.g. RADIUS) server, before submitting the username and password.

This certificate chain is received by the client's WiFi Supplicant, as part of the EAP authentication exchange.

The certificate chain is often not automatically verifiable, as it is self-signed, or uses a Certificate Authority (CA) that is not part of the user's trust root. Other times, the user connects to the WiFi network, and a captive portal is shown to them, (like in many airport or hotel WiFi networks) claiming to be the correct one, where the user should log in, and possibly pay for the connection.

Even though systems such as F-PKI [1] could mitigate some attacks, CAs cannot validate the geographic area which the WiFi network (represented by this SSID and certificate chain) covers, which renders the use of a CA moot. The use of captive portals many times provides a link between the domain name and the HTTPS connection, but not the network itself.

GeoPKI-GECKO [2] is a PKI that allows linking geographic positions from a Web certificate, as well as obtaining a set of Web certificates given a geographic location. By using GECKO, the user and WiFi network provider can guarantee that their WiFi network is listed as part of that geographical area, as well as provide an association between its certificates and its physical position.

This project task is to secure this important use case for Android devices, leveraging the GeoPKI-GECKO system [2].

Tasks

Design and implement an Android¹ application which interacts with one or several GECKO servers, and with the device WiFi connectivity [3].

1. Architect and design the application.
2. Design the graphical user interface (GUI) mockups.
3. In collaboration with the Behavioral Security and Privacy Group of the University Bonn [4], validate and improve the initial GUI and user interaction plan.
4. Design and implement the additional WiFi certificate verification .
5. Using the GECKO client API, design the security component that obtains the proper certificates from the geographic position. This certificate material might be obtained prior to the WiFi connection attempt (to be determined during the thesis).
6. Study the integration with payment web portals, such as those found at airports.
7. Study integrating GECKO into the system WiFi settings.
8. Test the approach with several WiFi deployments, specially with captive portals and EDUROAM.

¹ If the Proof of Concept is found to be unviable with Android, the fallback system would be Linux.

References

- [1] <https://netsec.ethz.ch/publications/papers/chuat2022fpki.pdf>
- [2] https://netsec.ethz.ch/publications/papers/2026_icnc_gecko.pdf
- [3] <https://developer.android.com/reference>
- [4] <https://www.besec.uni-bonn.de/team/team/matthew-smith>

Grading Scheme

Grade	Description
6.00	Design and implementation, as well as thesis are candidates for submission to an academic conference or workshop.
5.50	Thesis quality significantly exceeds expectations.
5.00	Thesis meets expectations.
4.50	Thesis partially meets expectations and has minor deficits.
4.00	Thesis meets minimum quality requirements; but has major deficits and is clearly below expectations

Date and Location:

Zürich, 19.08.26

Student's signature:



Advisor's signature:



Zürich, 7 Jul. 2026